



Responsible Artificial Intelligence Practices

Introduction

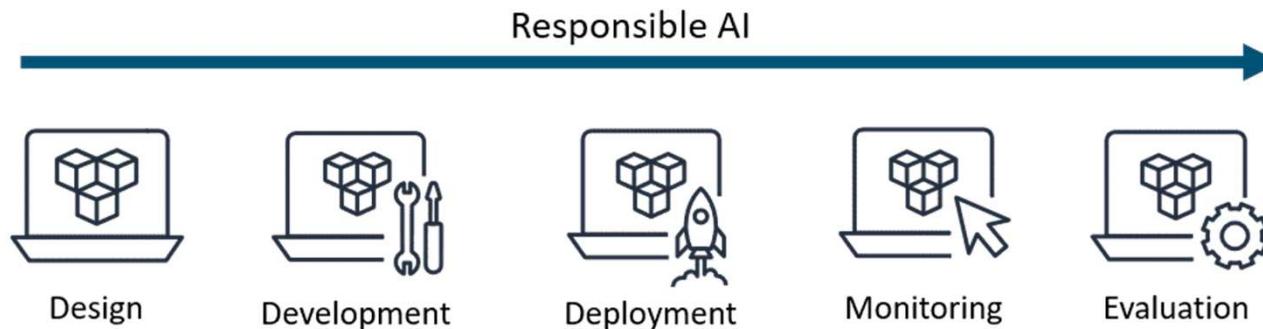
You will learn how to define responsible AI, understand the challenges that responsible AI attempts to overcome, and explore the core dimensions of responsible AI.

Dive into some topics for developing responsible AI systems. In this section of the course, you will learn about the services and tools that AWS offers to help you with responsible AI and tools involved

You will learn what it means for a model to be transparent and explainable.

What Is Responsible AI

Responsible AI refers to practices and principles that ensure that AI systems are transparent and trustworthy while mitigating potential risks and negative outcomes.



- It is fully transparent and accountable, with monitoring and oversight mechanisms in place.
- It is managed by a leadership team accountable for responsible AI strategies.
- It is developed by teams with expertise in responsible AI principles and practices.
- It is built following responsible AI guidelines.

Accuracy in AI Systems

It is important to address bias and variance in your model.

BIAS	VARIANCE
Bias arises when a model misses important data features, often due to overly simplistic or insufficient training data.	Variance refers to a model's sensitivity to noise in training data, where it may mistakenly treat noise as important features.
It is measured by the difference between predicted values and actual outcomes – small differences mean low bias; large differences mean high bias.	High variance leads to overfitting – excellent performance on training data but poor accuracy on new or unseen data.
High bias leads to underfitting, where the model performs poorly because it cannot capture data complexity.	Overfitting occurs because the model memorizes training data instead of learning patterns that generalize well.

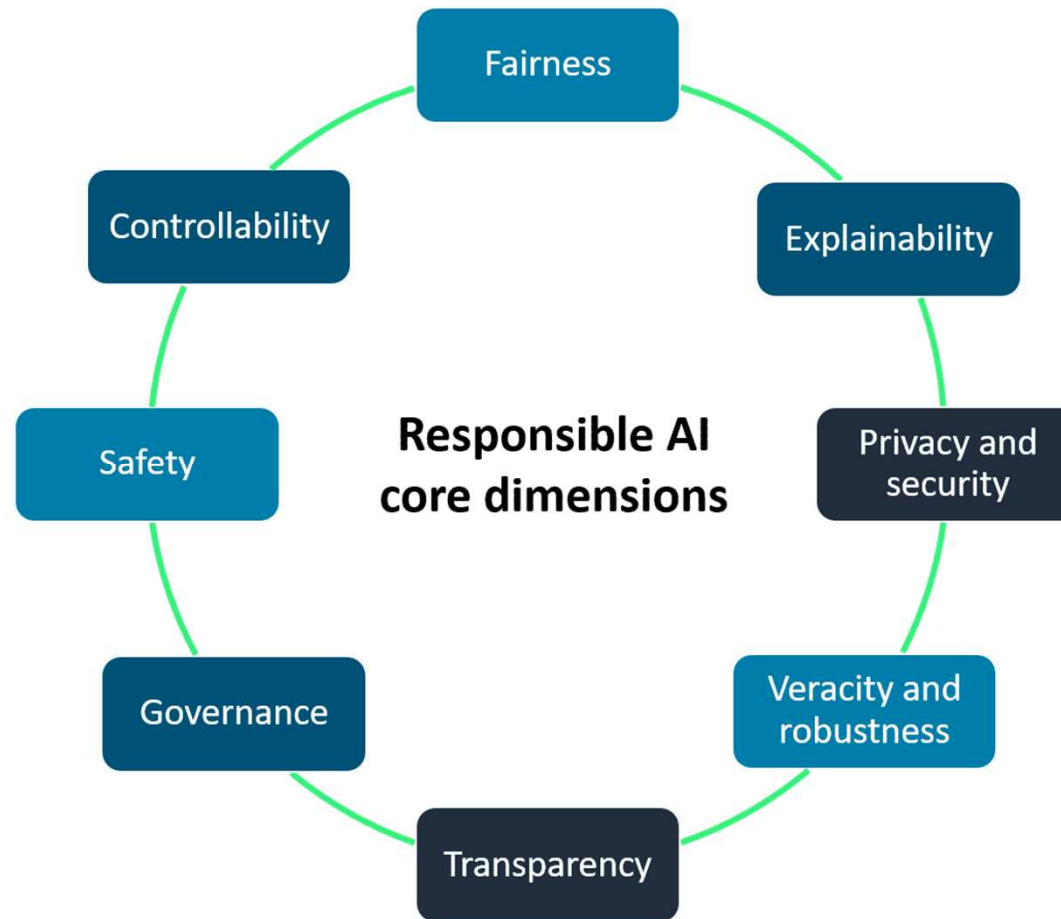
Overcoming Bias and Variance Errors

- Cross Validation
- Increase data
- Regularization
- Simpler Models
- Dimension Reduction (Principal Component Analysis)
- Stop training early

Challenges of Gen AI

- Toxicity
- Hallucinations
- Intellectual Property
- Plagiarism and Cheating
- Disruption of the Nature of Work

Core Dimensions of Responsible AI



Amazon Services and Tools

AWS offers services like Amazon SageMaker and Amazon Bedrock that have built-in tools to help you with responsible AI.

Amazon SageMaker	Amazon Bedrock
Data scientists and developers can quickly and confidently build, train, and deploy ML models into a production-ready hosted environment. It provides a UI experience for running ML workflows that makes SageMaker ML tools available across multiple integrated development environments (IDEs). Store and share your data without having to build and manage your own servers. Within a few steps, you can deploy a model into a secure and scalable environment from the SageMaker console.	A fully managed service with high-performing FMs from AI startups and Amazon for your use through a unified API. Amazon Bedrock also offers a broad set of capabilities to build generative AI applications with security, privacy, and responsible AI. With the serverless experience of Amazon Bedrock, customize FMs with your own data and securely integrate and deploy them into your applications by using AWS tools without having to manage any infrastructure.

Reviewing Amazon Services and Tools

Foundation model evaluation

- You should always evaluate a FM to determine if it will it is suited for your specific use case. To help you do this, Amazon offers model evaluation on Amazon Bedrock and Amazon SageMaker Clarify.

Amazon SageMaker	Amazon Bedrock
Amazon Bedrock offers a choice of automatic evaluation and human evaluation. Automatic evaluation offers predefined metrics such as accuracy, robustness, and toxicity. Human evaluation offers subjective or custom metrics such as friendliness, style, and alignment to brand voice. For human evaluation, you can use your in-house employees or an AWS-managed team as reviewers.	SageMaker Clarify supports FM evaluation. You can automatically evaluate FMs for your generative AI use case with metrics such as accuracy, robustness, and toxicity to support your responsible AI initiative. For criteria or nuanced content that requires sophisticated human judgment, you can choose to use your own workforce or use a managed workforce provided by AWS to review model responses.

Safeguards for generative AI

- With **Amazon Bedrock Guardrails**, you can implement safeguards for your generative AI applications based on your use cases and responsible AI policies.
- Guardrails control interaction between users and FMs by filtering undesirable and harmful content, redacting personally identifiable information (PII) and enhancing content safety and privacy in generative AI applications.
- Among other defined policies:
 - Consistent level of AI safety
 - Block undesirable topics
 - Filter harmful content
 - Redact PII to protect user privacy

Safeguards for generative AI

Bias Detection

SageMaker Clarify - identify potential bias in machine learning models and datasets without the need for extensive coding.

Amazon SageMaker Data Wrangler to balance your data in cases of any imbalances.

Model prediction explanation

- **SageMaker Clarify** is integrated with Amazon SageMaker Experiments to provide scores detailing which features contributed the most to your model prediction on a particular input for tabular, natural language processing (NLP), and computer vision models.

Safeguards for generative AI

Monitoring and human reviews

Amazon SageMaker Model Monitor monitors the quality of SageMaker machine learning models in production. You can set up continuous monitoring with a real-time endpoint (or a batch transform job that runs regularly), or on-schedule monitoring for asynchronous batch transform jobs. With SageMaker Model Monitor, you can set alerts that notify you when there are deviations in the model quality.

Amazon Augmented AI (Amazon A2I) is a service that helps build the workflows required for human review of ML predictions.

Safeguards for Generative AI

Governance tools include the following:

Amazon SageMaker Role Manager: With SageMaker Role Manager, administrators can define minimum permissions in minutes.

Amazon SageMaker Model Cards: With SageMaker Model Cards, you can capture, retrieve, and share essential model information, such as intended uses, risk ratings, and training details, from conception to deployment.

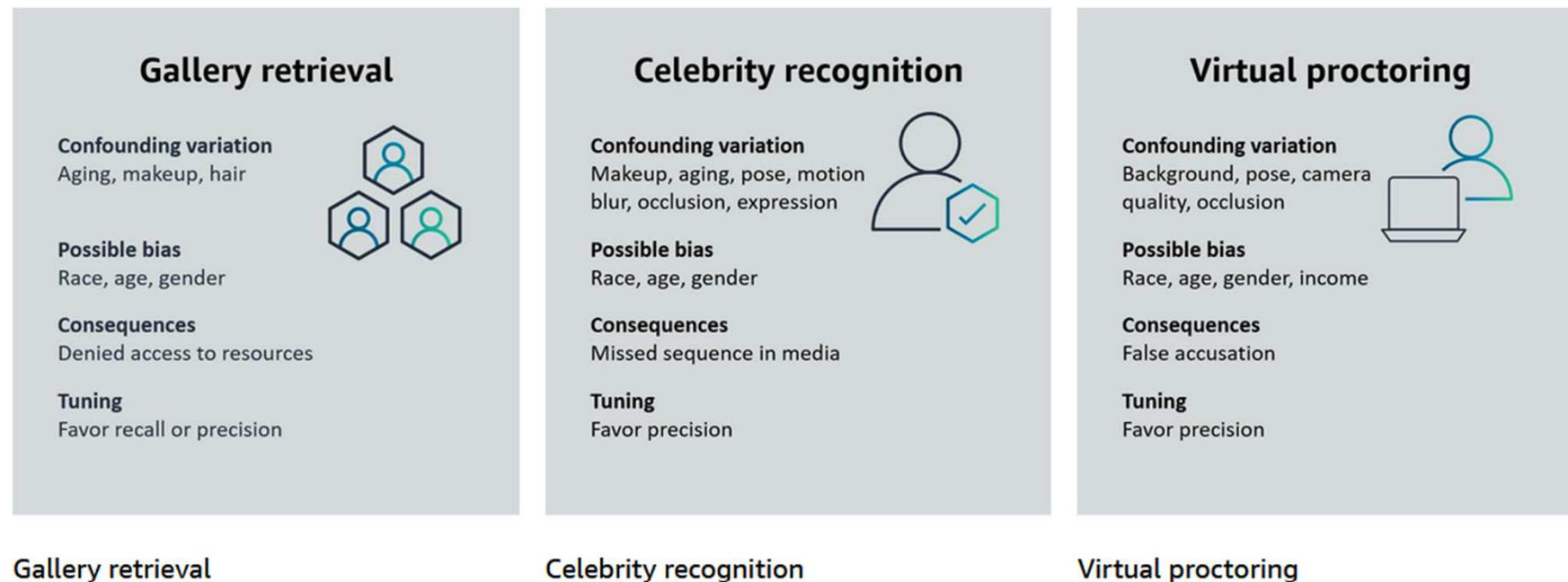
Amazon SageMaker Model Dashboard: With SageMaker Model Dashboard, you can keep your team informed on model behavior in production, all in one place.

Providing transparency

- **AWS AI Service Cards** are a new resource to help you better understand AWS AI services. AI Service Cards are a form of responsible AI documentation that provides a single place to find information on the intended use cases and limitations, responsible AI design choices, and deployment and performance optimization best practices for AWS AI services.

Responsible Considerations to Select a Model

Example: Defining application use case narrowly for traditional AI



Responsible Considerations to Select a Model

Example: Defining application use case narrowly for generative AI

Features	Catalog a product	Persuade to buy
Target audience	Broad demographic	Narrow demographic
Possible issues	Veracity	Veracity, unwanted bias, toxicity, detail
Consequences	Brand damage, lost sales, and returns	Representative harm, brand damage, lost sales, and returns
Tuning	Favors neutrality, clarity, and completeness	Focuses on highest interest problem and benefit to group

Responsible Preparation for Datasets

Amazon SageMaker Clarify and **SageMaker Data Wrangler** to help balance your dataset

- Balancing Datasets
 - Inclusive and diverse data collection



People data



Scientific data



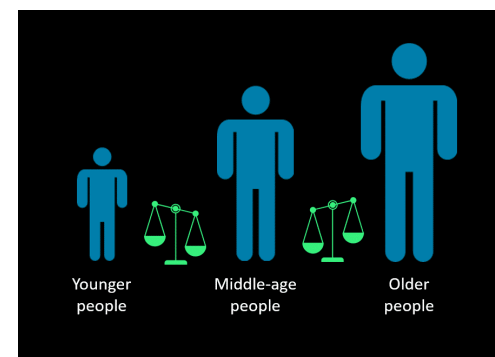
Geography data



Weather data



Product data



- Data curation and preparation

Data preprocessing

Data augmentation

Regular auditing

Transparent and Explainable Models

To promote trust and accountability in an AI system, there should be transparency and explainability in the model.

Transparency answers the question **HOW**, and explainability answers the question **WHY**. Both aspects are needed to build responsible AI systems.

- Increased trust
- Easier to debug and optimize for improvements
- Better understanding of the data and the model's decision-making process

AWS tools for transparency...

AWS AI Service Cards; Amazon provides transparent documentation on Amazon services that help you build your AI services.

With **SageMaker Model Cards**, you can catalog and provide documentation on models that you create or develop yourself.

and tools for explainability

SageMaker Clarify works with SageMaker Experiments to show which features most influence a model's prediction for individual inputs across tabular, NLP, and computer vision models. For tabular data, it also provides an aggregated feature importance chart to help identify if certain inputs have an outsized impact on the model's behavior.

SageMaker Autopilot uses tools provided by SageMaker Clarify to help provide insights into how ML models make predictions.

Model Trade-Offs

- Interpretability is a feature of model transparency. The access into a system so that a human can interpret the model's output based on the weights and features.
- Explainability is how to take an ML model and explain the behavior in human terms. With complex models (for example, black boxes), you cannot fully understand how and why the inner mechanics impact the prediction.

Model safety and model transparency trade-offs

- With model safety focusing on protecting information, and model transparency focusing on exposing information, you can understand that there is a delicate balance needed between them.
 - **Accuracy, Privacy, Safety and Security**

Knowledge Check.

Q1.

Responsible artificial intelligence (AI) should be incorporated in all phases of the development lifecycle of an AI system.

Which choice best describes responsible AI?

- A. Responsible AI refers the security and compliance that are shared responsibilities between AWS and the customer.
- B. Responsible AI refers to the standards that uphold responsible practices and mitigate potential risks and negative outcomes of an AI application.
- C. Responsible AI refers to the ability to create AI systems that add creativity, productivity, and connectivity to your business.
- D. Responsible AI refers to standards of upholding responsible practices that are exclusively needed for generative AI systems.

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Q2.

A developer is training a model for a new artificial intelligence (AI) system. The developer wants to be sure that the model is not underfitting the data or overfitting the data.

How should the model be trained on the data?

- A. With low bias and high variance
- B. With low bias and low variance
- C. With high bias and high variance
- D. With high bias and low variance

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A machine learning (ML) scientist is building an ML model for loan applications at a bank by using Amazon SageMaker. They want to mitigate bias in the data and increase visibility into model behavior.

- A. Amazon Comprehend
- B. Amazon SageMaker JumpStart
- C. Amazon Augmented AI (Amazon A2I)
- D. Amazon SageMaker Clarify

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Q4.

Monitoring is important to maintain high-quality machine learning (ML) models and help ensure accurate predictions.

Which AWS services or features help with monitoring and human review? (Select TWO.)

- A. Amazon SageMaker JumpStart
- B. Amazon SageMaker Model Monitor
- C. Amazon Augmented AI (Amazon A2I)
- D. Amazon SageMaker Clarify
- E. Amazon Bedrock

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Q5.

An organization is in the beginning stages of building an application that will use generative artificial intelligence (generative AI) technologies. The development team wants to use the best model for the application's use case. But with the large selection of large language models (LLMs), the development team is not sure which model to select. Also, the application will be public facing, and the team is concerned about the potential of generating harmful content.

Which AWS solutions should the team use to address these concerns? (Select TWO)

- A. Amazon SageMaker Model Dashboard
- B. Model Evaluation on Amazon Bedrock
- C. Amazon Augmented AI (Amazon A2I)
- D. Amazon SageMaker Data Wrangler
- E. Amazon Bedrock Guardrails

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